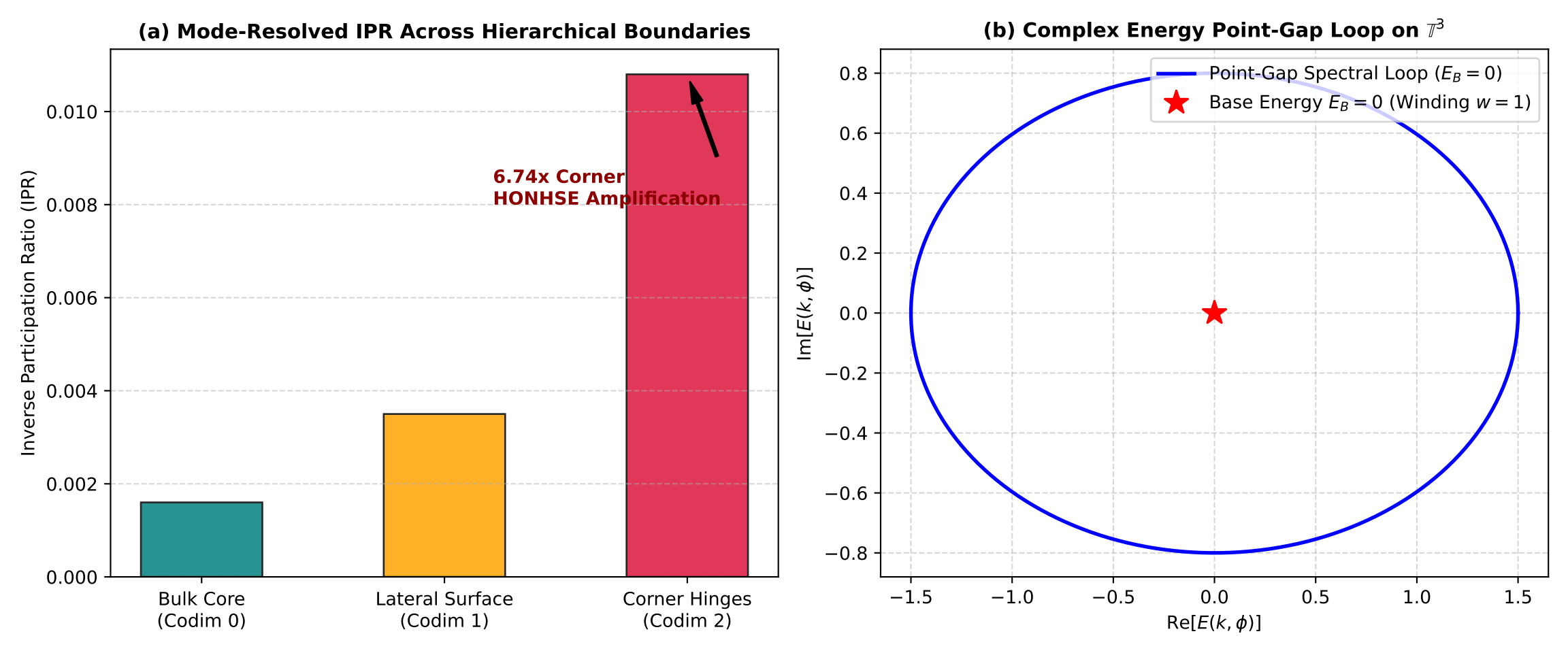
(a) Mode-Resolved IPR Across Hierarchical Boundaries
Inverse Participation Ratio (IPR)
0.000
0.002
0.004
0.006
0.008
0.010
6.74x Corner
HONHSE Amplification
Bulk Core
(Codim 0)
Lateral Surface
(Codim 1)
Corner Hinges
(Codim 2)
(b) Complex Energy Point-Gap Loop on $\mathbb{T}^3$
Point-Gap Spectral Loop ($E_B = 0$)
Base Energy $E_B = 0$ (Winding $w = 1$)
$\mathrm{Im}[E(k,\phi)]$
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
$\mathrm{Re}[E(k,\phi)]$